

HTHHT

$$S \rightarrow \underline{HT} \rightarrow HT \underline{HT}$$

4 32

$$1E[HTHHT] = 36$$

"SANITY CHECK"

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \\ E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTH} \\ E_{HTH} = 1 + \frac{1}{2}E_{HT} + \frac{1}{2}E_{HTHH} \\ E_{HTHH} = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HTHHT} \\ E_{HTHHT} = 0 \end{cases}$$

$$E_{HTHH} = 1 + \frac{1}{2}E_H$$

$$\begin{aligned} E_{HTH} &= 1 + \frac{1}{2}E_{HT} + \frac{1}{2}\left(1 + \frac{1}{2}E_H\right) \\ &= \frac{3}{2} + \frac{1}{2}E_{HT} + \frac{1}{4}E_H \end{aligned}$$

$$E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{2}E_{HT} + \frac{1}{4}E_H\right)$$

$$\frac{3}{4}E_{HT} = \frac{7}{4} + \frac{1}{2}E_S + \frac{1}{8}E_H$$

$$E_{HT} = \frac{7}{3} + \frac{2}{3}E_S + \frac{1}{6}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}\left(\frac{7}{3} + \frac{2}{3}E_S + \frac{1}{6}E_H\right)$$

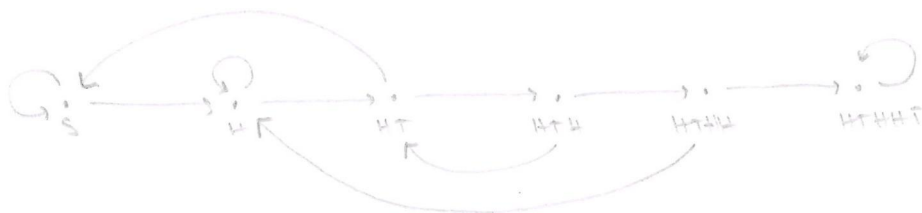
$$\frac{5}{12}E_H = \frac{13}{6} + \frac{1}{3}E_S$$

$$E_H = \frac{26}{5} + \frac{4}{5}E_S$$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{26}{5} + \frac{4}{5}E_S\right)$$

$$\frac{1}{10}E_S = \frac{18}{5}$$

$$E_S = 36$$



HTTTHT

$$S \rightarrow \underline{HT} \rightarrow \text{HTTT}\underline{HT} \quad \text{IE } [HTTTHT] = 4 + 64 = 68$$

IDEA : what is the longest bit of sequence (sub-sequence) that will allow me to obtain again the entire sequence?

if I have HT then I need "only" TTHT again to obtain the full sequence again

HTTTHT TTHT

"SANITY CHECK":



ALL PROB = $\frac{1}{2}$

E_x = number of flips from x to HTTTHT

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$$E_{HT} = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HTT}$$

$$E_{HTT} = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HTTT}$$

$$E_{HTTT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTTTHT}$$

$$E_{HTTTHT} = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HTTTHTT}$$

$$E_{HTTTHTT} = 0$$

$$E_{HTTTHTT} = 1 + \frac{1}{2}E_H$$

$$\begin{aligned} E_{HTTT} &= 1 + \frac{1}{2}E_S + \frac{1}{2}\left(1 + \frac{1}{2}E_H\right) \\ &= \frac{3}{2} + \frac{1}{2}E_S + \frac{1}{4}E_H \end{aligned}$$

$$\begin{aligned} E_{HTT} &= 1 + \frac{1}{2}E_H + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{2}E_S + \frac{1}{4}E_H\right) \\ &= \frac{7}{4} + \frac{1}{4}E_S + \frac{5}{8}E_H \end{aligned}$$

$$\begin{aligned} E_{HT} &= 1 + \frac{1}{2}E_H + \frac{1}{2}\left(\frac{7}{4} + \frac{1}{4}E_S + \frac{5}{8}E_H\right) \\ &= \frac{15}{8} + \frac{1}{8}E_S + \frac{13}{16}E_H \end{aligned}$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}\left(\frac{15}{8} + \frac{1}{8}E_S + \frac{13}{16}E_H\right)$$

$$\frac{3}{32}E_H = \frac{31}{16} + \frac{1}{16}E_S$$

$$E_H = \frac{62}{3} + \frac{2}{3}E_S$$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{62}{3} + \frac{2}{3}E_S\right)$$

$$\frac{1}{6}E_S = \frac{34}{3}$$

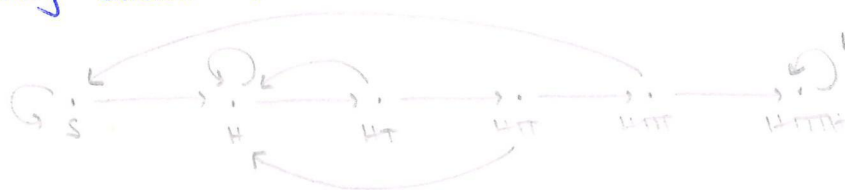
$$E_S = 68 \quad \text{!!}$$

HTTTH

$$S \xrightarrow[2]{H} \xrightarrow[32]{HTTTH}$$

$$1E[HTTTH] = 34$$

"SANITY CHECK":



$$\text{ALL PROB} = \frac{1}{2}$$

E_x = number of steps from x to HTTTH

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \\ E_{HT} = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HTT} \\ E_{HTT} = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HTTTH} \\ E_{HTTTH} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTTTH} \\ E_{HTTTH} = 0 \end{cases}$$

$$E_{HTTTH} = 1 + \frac{1}{2}E_S$$

$$\begin{aligned} E_{HTT} &= 1 + \frac{1}{2}E_H + \frac{1}{2}\left(1 + \frac{1}{2}E_S\right) \\ &= \frac{3}{2} + \frac{1}{2}E_H + \frac{1}{4}E_S \end{aligned}$$

$$\begin{aligned} E_{HT} &= 1 + \frac{1}{2}E_H + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{2}E_H + \frac{1}{4}E_S\right) \\ &= 1 + \frac{1}{2}E_H + \frac{3}{4} + \frac{1}{4}E_H + \frac{1}{8}E_S \\ &= \frac{7}{4} + \frac{3}{4}E_H + \frac{1}{8}E_S \end{aligned}$$

$$\begin{aligned} E_H &= 1 + \frac{1}{2}E_H + \frac{1}{2}\left(\frac{7}{4} + \frac{3}{4}E_H + \frac{1}{8}E_S\right) \\ &= 1 + \frac{1}{2}E_H + \frac{7}{8} + \frac{3}{8}E_H + \frac{1}{16}E_S \end{aligned}$$

$$\frac{1}{8}E_H = \frac{15}{8} + \frac{1}{16}E_S$$

$$E_H = 15 + \frac{1}{2}E_S$$

$$\begin{aligned} E_S &= 1 + \frac{1}{2}E_S + \frac{1}{2}\left(15 + \frac{1}{2}E_S\right) \\ &= 1 + \frac{1}{2}E_S + \frac{15}{2} + \frac{1}{4}E_S \end{aligned}$$

$$\frac{1}{4}E_S = \frac{17}{2} \quad E_S = 34 \quad \text{II}$$

COMMENTS ON "SIMILAR" SEQUENCES

HTHT : $S \rightarrow \underline{HT} \rightarrow \underline{HTHT}$ $IE[HTHT] = 20$
HT REAPPEARS as a WHOLE to OBTAIN
THE ENTIRE SEQUENCE AGAIN

HTHTH : $S \rightarrow \underline{H} \rightarrow \underline{HTH} \rightarrow \underline{HTHTH}$ $IE[HTHTH] = 42$
TO OBTAIN THE ENTIRE SEQUENCE WE NEED TO
CHECK FIRST WHEN H REAPPEARS (ALSO TO
"CLOSE" THE SEQUENCE WHICH ENDS WITH
H - NOT HT)

HTHTHT : $S \rightarrow \underline{HT} \rightarrow \underline{HTHT} \rightarrow \underline{HTHTHT}$ $IE[HTHTHT] = 84$

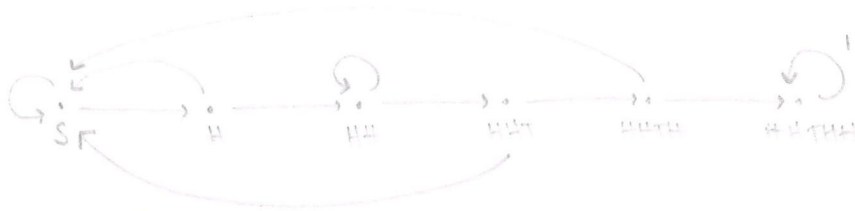
HTHTHTHT : $S \rightarrow \underline{HT} \rightarrow \underline{HTHT} \rightarrow \underline{HTHTHT} \rightarrow \underline{HTHTHTHT}$
WE HAVE TO DO ALL THE STEPS BECAUSE HT
REPEATS ITSELF ALL THE TIME TILL THE END
AND EACH OF THEM CAN BE RE-USED
TO OBTAIN AGAIN THE ENTIRE SEQUENCE

HTTHH

$$S \xrightarrow{2} H \xrightarrow{4} HH \xrightarrow{32} HTTHH$$

$$IE[HTTHH] = 38$$

"SANITY CHECK":



ALL PROB = $\frac{1}{2}$

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH} \\ E_{HH} = 1 + \frac{1}{2}E_{HH} + \frac{1}{2}E_{HHT} \\ E_{HHT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTTH} \\ E_{HTTH} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTTHH} \\ E_{HTTHH} = 0 \end{cases}$$

E_x = number of flips from x to HTTHH

$$E_{HTTHH} = 1 + \frac{1}{2}E_S$$

$$\begin{aligned} E_{HHT} &= 1 + \frac{1}{2}E_S + \frac{1}{2}\left(1 + \frac{1}{2}E_S\right) \\ &= \frac{3}{2} + \frac{3}{4}E_S \end{aligned}$$

$$E_{HH} = 1 + \frac{1}{2}E_{HHT} + \frac{1}{2}\left(\frac{3}{2} + \frac{3}{4}E_S\right)$$

$$\frac{1}{2}E_{HH} = 1 + \frac{3}{4} + \frac{3}{8}E_S$$

$$E_{HH} = 2 + \frac{3}{2} + \frac{3}{4}E_S$$

$$= \frac{7}{2} + \frac{3}{4}E_S$$

$$E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{7}{2} + \frac{3}{4}E_S\right)$$

$$= 1 + \frac{1}{2}E_S + \frac{7}{4} + \frac{3}{8}E_S$$

$$= \frac{11}{4} + \frac{7}{8}E_S$$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{11}{4} + \frac{7}{8}E_S\right)$$

$$= 1 + \frac{1}{2}E_S + \frac{11}{8} + \frac{7}{16}E_S$$

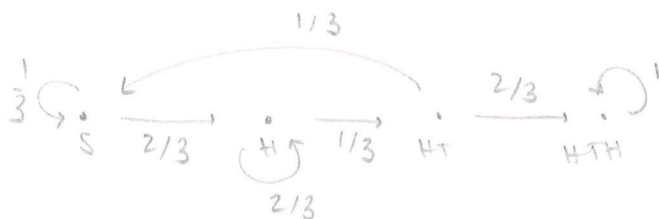
$$\frac{1}{16}E_S = \frac{19}{8}$$

$$E_S = 38 \quad \text{||}$$

UNFAIR COIN

$$IP(H) = \frac{2}{3} \quad IP(T) = \frac{1}{3}$$

What is the expected number of flips until HTH appears for the first time?



E_x = number of flips to go from state x to HTH

$$\begin{cases} E_S = 1 + \frac{1}{3}E_S + \frac{2}{3}E_H \\ E_H = 1 + \frac{1}{3}E_{HT} + \frac{2}{3}E_{HTH} \\ E_{HT} = 1 + \frac{2}{3}E_{HTH} + \frac{1}{3}E_S \\ E_{HTH} = 0 \end{cases}$$

$$E_{HT} = 1 + \frac{1}{3}E_S$$

$$E_H = 1 + \frac{1}{3}\left(1 + \frac{1}{3}E_S\right) + \frac{2}{3}E_{HTH}$$

$$\frac{1}{3}E_H = 1 + \frac{1}{3} + \frac{1}{9}E_S$$

$$= \frac{4}{3} + \frac{1}{9}E_S$$

$$E_H = 4 + \frac{1}{3}E_S$$

$$E_S = 1 + \frac{1}{3}E_S + \frac{2}{3}\left(4 + \frac{1}{3}E_S\right)$$

$$= 1 + \frac{1}{3}E_S + \frac{8}{3} + \frac{2}{9}E_S$$

$$\left(1 - \frac{1}{3} - \frac{2}{9}\right)E_S = \frac{11}{3}$$

$$\frac{4}{9}E_S = \frac{11}{3}$$

$$E_S = \frac{11}{3} \cdot \frac{9}{4} = \frac{33}{4}$$

FUPs HHT with UNFAIR COIN

$$P(H) = \frac{2}{3}$$

$$P(T) = \frac{1}{3}$$

$$\begin{cases} E_S = 1 + \frac{1}{3} E_S + \frac{2}{3} E_H \\ E_H = 1 + \frac{1}{3} E_S + \frac{2}{3} E_{HH} \\ E_{HH} = 1 + \frac{2}{3} E_{HH} + \frac{1}{3} E_{HHT} \\ E_{HHT} = 0 \end{cases}$$

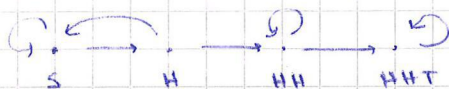
$$E_{HH} = 3$$

$$\begin{aligned} E_S &= 1 + \frac{1}{3} E_S + \frac{2}{3} \left(1 + \frac{1}{3} E_S + 2 \right) \\ &= 1 + \frac{1}{3} E_S + 2 + \frac{2}{9} E_S \end{aligned}$$

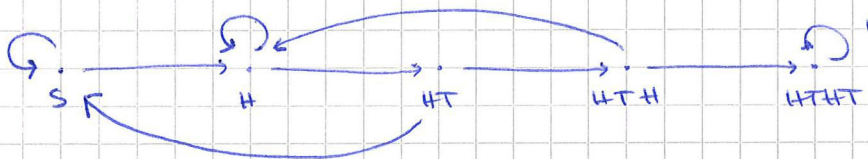
$$\left(1 - \frac{1}{3} - \frac{2}{9} \right) E_S = 3$$

$$\frac{9 - 3 - 2}{9} E_S = 3$$

$$E_S = \frac{27}{4}$$



FUPs HTHT - FAIR COIN



ALL PROBS = $\frac{1}{2}$

$$S \xrightarrow{H} HT \xrightarrow{T} HTHT \quad E_S = 20$$

$$\begin{cases} E_S = 1 + \frac{1}{2} E_S + \frac{1}{2} E_H \\ E_H = 1 + \frac{1}{2} E_H + \frac{1}{2} E_{HT} \\ E_{HT} = 1 + \frac{1}{2} E_S + \frac{1}{2} E_{HTH} \\ E_{HTH} = 1 + \frac{1}{2} E_H + \frac{1}{2} E_{HTHT} \\ E_{HTHT} = 0 \end{cases}$$

$$E_{HTH} = 1 + \frac{1}{2} E_H$$

$$\begin{aligned} E_{HT} &= 1 + \frac{1}{2} E_S + \frac{1}{2} \left(1 + \frac{1}{2} E_H \right) \\ &= \frac{3}{2} + \frac{1}{2} E_S + \frac{1}{4} E_H \end{aligned}$$

$$\begin{aligned} E_H &= 1 + \frac{1}{2} E_H + \frac{1}{2} \left(\frac{3}{2} + \frac{1}{2} E_S + \frac{1}{4} E_H \right) \\ &= \frac{7}{4} + \frac{1}{2} E_H + \frac{1}{8} E_H + \frac{1}{4} E_S \end{aligned}$$

$$E_H = \frac{8}{3} \left(\frac{7}{4} + \frac{1}{4} E_S \right) = \frac{14}{3} + \frac{2}{3} E_S$$

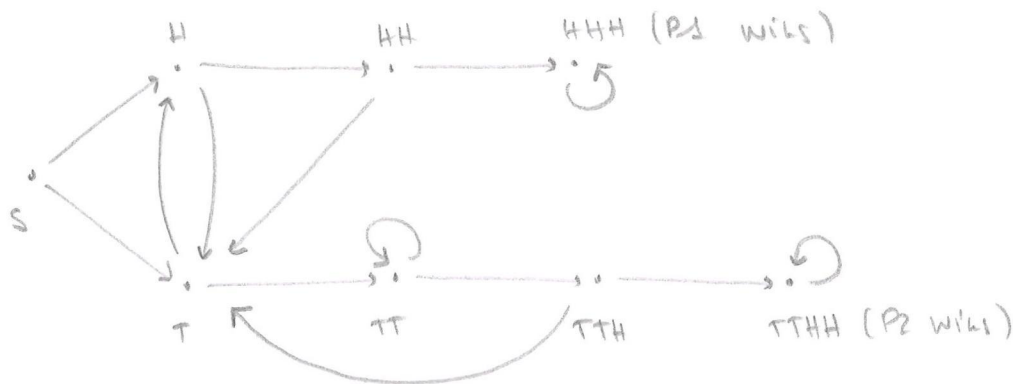
$$E_S = 1 + \frac{1}{2} E_S + \frac{1}{2} \left(\frac{14}{3} + \frac{2}{3} E_S \right) = 1 + \frac{1}{2} E_S + \frac{7}{3} + \frac{1}{3} E_S \quad E_S = 20$$

Imagine you have a fair coin, which you toss repeatedly.
Two players observe the sequence of tosses.

P1 waits for HHH

P2 waits for TTH

the one whose pattern appear first is the winner
What is the probability that P1 wins?



P_x = prob of P1 winning

$$P_S = \frac{1}{2} P_H + \frac{1}{2} P_T$$

$$P_H = \frac{1}{2} P_{HH} + \frac{1}{2} P_T$$

$$P_T = \frac{1}{2} P_{TT} + \frac{1}{2} P_H$$

$$P_{HH} = \frac{1}{2} P_T + \frac{1}{2} P_{HHH}$$

$$P_{TT} = \frac{1}{2} P_{TTT} + \frac{1}{2} P_{TTH}$$

$$P_{TTH} = \frac{1}{2} P_T + \frac{1}{2} P_{TTTH}$$

$$P_{HHH} = 1$$

$$P_{TTTH} = 0$$

$$P_{TTTH} = 0$$

$$P_{HHH} = 1$$

$$P_{TTH} = \frac{1}{2} P_T$$

$$P_{TT} = \frac{1}{2} P_{TTT} + \frac{1}{2} \left(\frac{1}{2} P_T \right)$$

$$\frac{1}{2} P_{TT} = \frac{1}{2} P_T$$

$$P_{TT} = \frac{1}{2} P_T$$

$$P_{HH} = \frac{1}{2} P_T + \frac{1}{2} P_{HHH} = \frac{1}{2} P_T + \frac{1}{2}$$

$$P_T = \frac{1}{2} P_{TT} + \frac{1}{2} P_H$$

$$= \frac{1}{2} \left(\frac{1}{2} P_T \right) + \frac{1}{2} P_H = \frac{1}{4} P_T + \frac{1}{2} P_H$$

$$\frac{3}{4} P_T = \frac{1}{2} P_H$$

$$P_T = \frac{2}{3} P_H$$

$$P_H = \frac{1}{2} P_{HH} + \frac{1}{2} P_T$$

$$= \frac{1}{2} \left(\frac{1}{2} P_T + \frac{1}{2} \right) + \frac{1}{2} P_T$$

$$= \frac{3}{4} P_T + \frac{1}{4} = \frac{3}{4} \cdot \frac{2}{3} P_H + \frac{1}{4}$$

$$\frac{1}{2} P_H = \frac{1}{4}$$

$$P_H = \frac{1}{2}$$

$$P_S = \frac{1}{2} P_H + \frac{1}{2} P_T$$

$$= \frac{1}{2} P_H + \frac{1}{3} P_H$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$